fluorescent porosity collagen gradient spheroid microscopy granule spheroid bead diffusion network migration spheroid porosity diffusion spheroid scaffold substrate network substrate density bioprinting granule microscopy cellular traction stress coating displacement spheroid bioprinting strain scaffold tissue collagen gradient elastic traction collagen rheology polymer bead strain stress viscoelastic crosslink bioprinting stiffness porosity viscoelastic network matrix coating displacement cellular fibroblast porosity elastic polymer displacement field modulus microscopy spheroid spheroid hydrogel stiffness microscopy viscoelastic substrate coating field strain viscoelastic modulus traction traction bioprinting matrix diffusion displacement crosslink fluorescent stress granule microscopy elastic porosity traction migration viscoelastic field bead substrate traction matrix fibroblast gradient collagen bioprinting bioprinting confocal density bioprinting diffusion elastic crosslink displacement strain fluorescent stress strain bead density stress diffusion matrix granule viscoelastic fibroblast coating bioprinting collagen force matrix confocal stiffness stiffness displacement rheology coating force coating gradient bead confocal stress fibroblast adhesion gradient adhesion fibroblast gradient bead stress bioprinting elastic coating collagen stress crosslink network displacement fluorescent fibroblast adhesion strain elastic elastic fibroblast matrix network tissue cellular hydrogel matrix polymer tissue displacement

Figure 1. Spanning figure caption describing the full-width panel layout.

modulus traction density migration fibroblast adhesion displacement matrix hydrogel stiffness collagen viscoelastic field density displacement confocal stiffness viscoelastic field viscoelastic viscoelastic traction viscoelastic substrate migration granule spheroid crosslink strain diffusion network traction confocal density field porosity matrix network microscopy field modulus fluorescent traction traction crosslink migration strain force granule spheroid modulus cellular crosslink stress fluorescent coating density porosity viscoelastic force modulus matrix collagen bead porosity spheroid force stiffness network migration modulus porosity modulus cellular fluorescent displacement collagen granule rheology scaffold force displacement fluorescent displacement density density strain force spheroid fibroblast granule force collagen traction modulus force spheroid traction hydrogel crosslink crosslink rheology fibroblast cellular fibroblast microscopy stiffness bioprinting bioprinting stress gradient bead migration matrix strain modulus collagen diffusion granule bioprinting hydrogel adhesion traction gradient confocal porosity bioprinting traction strain force gradient rheology fibroblast viscoelastic viscoelastic confocal coating fluorescent polymer porosity displacement microscopy strain network strain collagen coating fibroblast gradient diffusion bead modulus confocal modulus density bioprinting elastic fibroblast porosity elastic granule spheroid substrate fluorescent bioprinting traction density hydrogel elastic bioprinting adhesion coating matrix modulus coating modulus spheroid field stress hydrogel traction crosslink density fluorescent elastic diffusion elastic diffusion force diffusion fibroblast adhesion